Worksight · Desert Eagles

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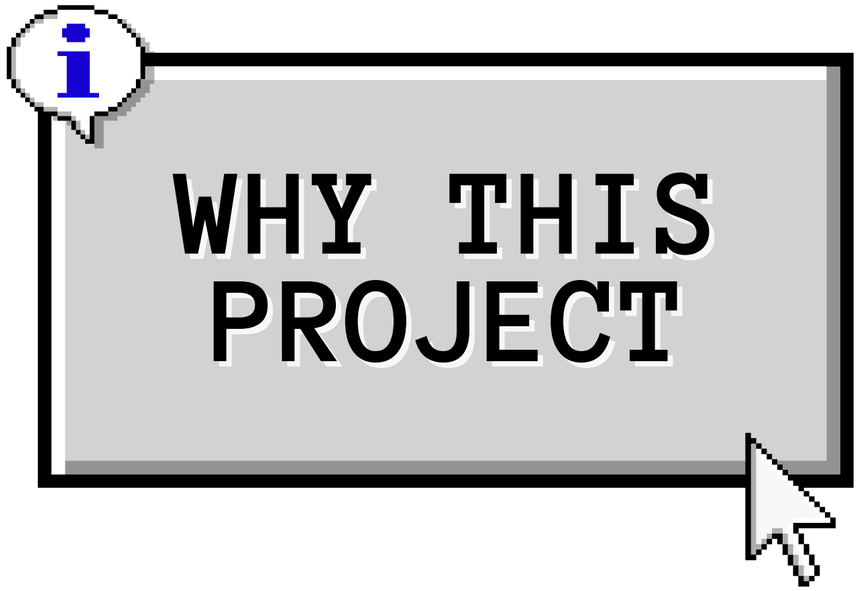
AMD Developer Hackathon · ACT III

WORKSIGHT

DEFECTS FROM GOOD PHOTOS



Start

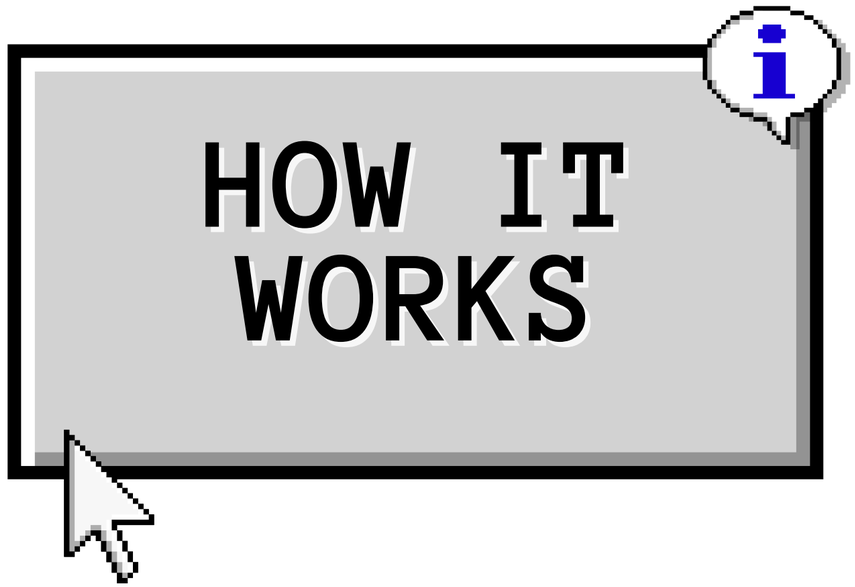


A new part on the line has no defect dataset. Defects are rare, so collecting and labelling hundreds of them takes weeks, and every new part or camera view starts again. Worksight starts from what every line already has: photos of good parts.

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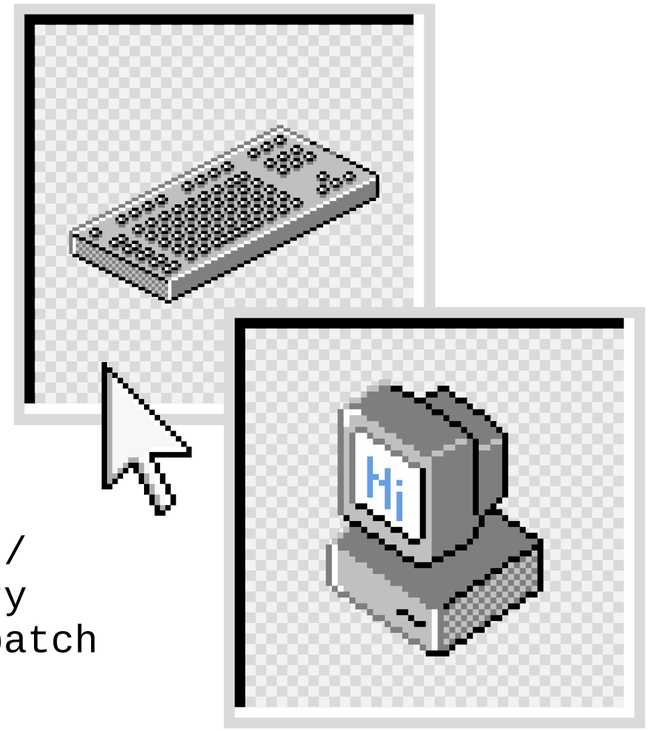
1. 20 photos of good parts build a profile in about 9 s. No labels, no training.
2. Each part gets an anomaly score, a heatmap and the closest good reference.
3. Gemma on AMD explains what to check. The engineer decides.



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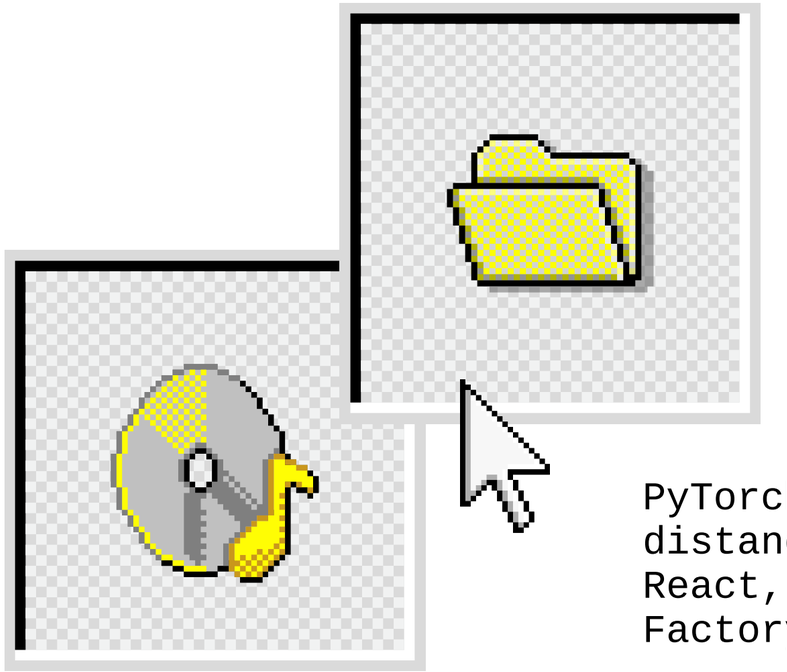
THE LIVE PRODUCT



Algorithm, AI and engineer on one screen: heatmap and closest reference, Pass / Fail / Needs review, Gemma's explanation with every inspection, one-page PDF report, history, batch inspection, 7 languages.

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TECH STACK

PyTorch: frozen ResNet-18 features, nearest-patch distance. FastAPI, SQLite, job queue. Next.js 16, React, TypeScript. Gemma-4-31B-IT on AMD Token Factory. Railway. pytest, Vitest, Playwright.

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BENCHMARKS

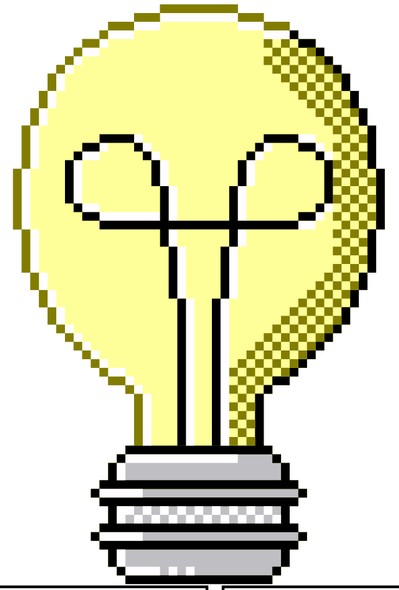
VisA PCB · held-out test · 20 good references · CPU

Set	AUROC	F1	Recall	False alarms	Defects to engineer	Good to engineer	p50
pcb1	0.905	0.591	44%	5%	**56%**	7%	250 ms
pcb2	0.884	0.715	59%	6%	**77%**	13%	304 ms
pcb3	0.875	0.613	46%	4%	**58%**	10%	250 ms
pcb4	0.992	0.921	87%	2%	**96%**	4%	194 ms

Recall = defects above the threshold. "To engineer" counts the review band (within 10% of the threshold). Mean AUROC 0.914. CPU run, not an AMD GPU. Source: docs/BENCHMARK.md

PARTNER TECH & WHY

AMD Token Factory runs Gemma on AMD GPUs; the API key stays on our server. Gemma 4 31B reads the photo, heatmap, reference and part documents and explains them in plain words. DeepSeek V4.1 Flash is the fallback. Next: the detector on AMD Instinct with ROCm.



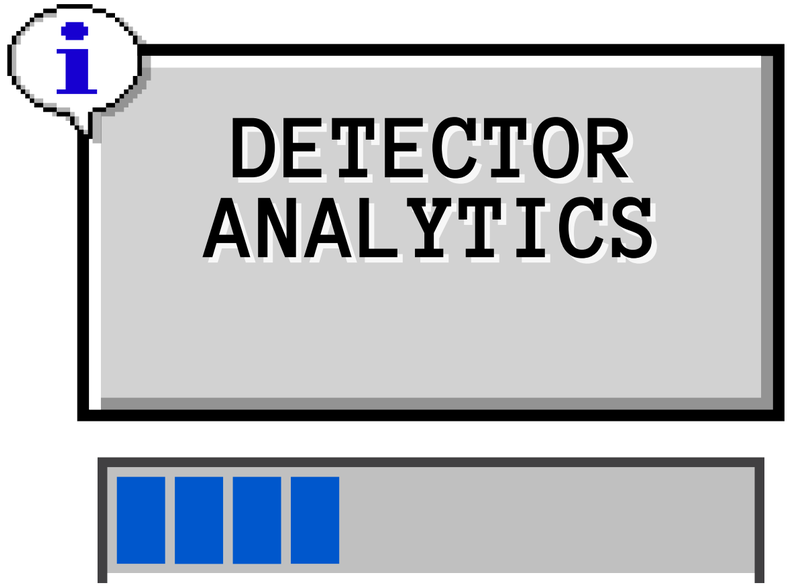
Hackathon partners, sponsors and venues

AMD · Gemma 4 · LAB LAB · evolus · IMMENT STARTUP CAPITAL ENGINE · UNIVERSITÀ DEGLI STUDI LINK · SMARTCITYLAB MILANO · imperiaware polo digitale imperia

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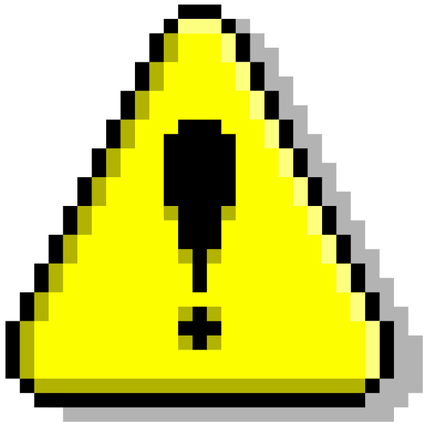
AUROC 0.88 to 0.99 on four circuit-board sets. F1 0.59 to 0.92.



v2 raised dev AUROC from 0.806 to 0.862. A 24-variant sweep found nothing better, so production stayed unchanged.

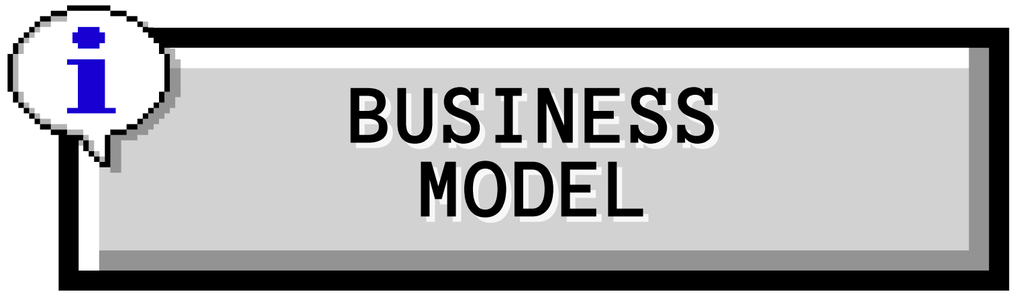
The engineer decides; model and engineer decisions are stored separately. Gemma never changes the detector result. A new profile version goes live only if it adds no false alarm or missed defect, and it can be rolled back.







AI visual inspection is forecast to grow by about $27B by 2029 (Technavio). Most tools need labelled defects or are code libraries. Worksight starts from good photos, explains each result and keeps the engineer in charge.



BUSINESS MODEL

Yes

Cancel

Subscription per camera view, or on-premises on AMD hardware for plants that keep images on site. Pilot on one line and one camera: measure engineer minutes per part and escaped defects.

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Bakhtiyar, CEO and AI engineer: detector, benchmarks, Gemma integration. Ali, frontend developer: operator workspace and reports. Next: detector on AMD Instinct, team workspaces, edge station on AMD Ryzen AI.



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